
CivicCast User Manual

For station operators, clerks, and IT staff · v1.0.0-rc18 release candidate

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2026-07-21

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CivicCast User Manual

CivicCast is open-source, self-hostable civic meeting recording and publication software. This controlled beta proves a bounded local path: create or upload recorded media, rehearse that exact sample privately, package it privately, explicitly approve Portal publication, and play it in the resident portal. Live ingest, 24/7 station operation, hardware/headend delivery, external providers, and app stores need separate integration and field proof.

This manual is one document in three sections, each pitched at a different reader:

- **Section A — End-User Guide.** For board members, clerks’ office staff, meeting operators, and anyone who is going to *use* CivicCast without needing to maintain it. No jargon. If a technical term is unavoidable, it is defined inline.
- **Section B — Technical Reference.** For the IT staff, integrators, and ops people who install, configure, monitor, and troubleshoot the station. Includes the CLI, the environment-variable surface, the credential store, and the role and permission model.
- **Section C — Architecture Reference.** For developers, auditors, and technical reviewers. Migration chain, the GStreamer playout engine, the three Protocol seams that connect the new modules to capture/publish/ alerting, the CDN-aware proxy resolver, and pointers into the v3.0 spec set.

The source inventory includes software-side PEG station-in-a-box capabilities plus control-room and virtual media studio beta work (see [Comparative Capability Status](#) in Section C). These source capabilities and their lab evidence are not stock acceptance claims and do not establish station-device, provider, app-store, or production proof. The rc13 Windows installer is withdrawn after a genuine clean-host bootstrap failure. Do not install rc13. v1.0.0-rc18 is the published controlled beta: its installer is built from the gate-cleared main, Authenticode-signed, and proven on a genuinely clean Windows host through install, launch, reinstall, uninstall and the rc17 upgrade.

Who Reads What

Reader	Start In	Then See
First-time user / board member / clerk	Section A	Meeting Operator Guide , Records Clerk Guide
Station admin / IT lead	Section B	Admin Guide , Technical Operations Reference

Reader	Start In	Then See
Developer / integrator / auditor	Section C	docs/spec/3.0/RECONCILIATION.md

Section A — End-User Guide

This section is written for people who do not live in terminals. If you can operate a meeting, post a flyer to a community board, or upload a video to a website, you can run CivicCast.

If something here points you to a screen you do not have, ask your IT lead or station admin. The page they need is in [Section B — Technical Reference](#).

What Is CivicCast?

CivicCast’s broader source inventory contains work toward the following product directions. These are not all stock acceptance claims:

- **Integrated live meetings.** With a separately proven server-side media probe (the automated check that confirms a live signal is actually present before **Start Live Stream** can turn on) and station egress (the station’s own connection that sends the stream out to viewers), a council meeting can stream to residents in real time on the public website, on the local cable channel, and (if the station turns it on) through station-operated web or OTT endpoints (apps on devices like Roku or Apple TV). Publication through commercial app stores requires provider accounts and store review and is outside this package.
- **Integrated 24/7 cable operation.** Past meetings, a community-bulletin board, and short underwriting spots can be scheduled to fill the channel between live events — without anyone in the studio.
- **Publish the record.** The source includes captions, summaries, signed PDF transcripts, and chapter markers tied to the meeting agenda. Each station must separately prove and configure the optional paths it intends to use.
- **Reach residents who do not watch live.** Subscribers get emails, a podcast feed, or app notifications when a new meeting is published.
- **Stay accountable.** Every published recording carries a tamper-evident trust stamp (an RFC 3161 timestamp). The signed PDF transcript is a review artifact; each jurisdiction decides what constitutes its legal record.

CivicCast runs on your hardware. Stock acceptance keeps the tested media local. Optional surfaces — Internet Archive mirroring, YouTube simulcast, emergency-alert ingest — turn on per-station, with the station’s own accounts.

Your First Beta Workflow

This walkthrough assumes someone has finished the installer and you have a username and password for the operator console.

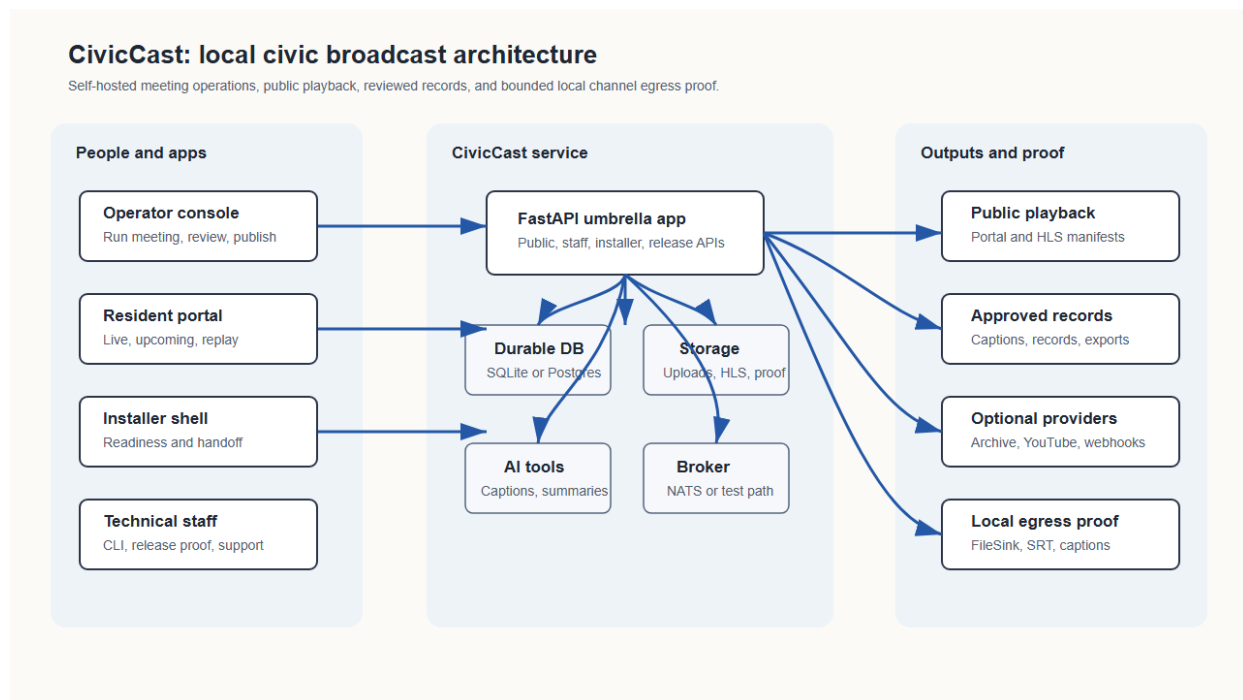


Figure 1: CivicCast system architecture

Beta boundary. The stock build has no production server-side media probe. It shows **Source preview unavailable** and keeps **Start Live Stream** disabled. That is the expected safe state, not an installation failure.

1. **Open the operator console.** Your IT lead will tell you the address (it usually looks like <https://broadcast.yourtown.gov>). Sign in.
2. **Confirm the live safety state.** Open **Run Meeting** and confirm the room says **Source preview unavailable** and does not enable live start.
3. **Create or upload sample recorded media.** Use non-sensitive test content.
4. **Run a private rehearsal.** In **System Health**, choose **Run private rehearsal**. CivicCast must report that it copied the exact validated sample, created and finalized a private recording, and loaded resident preview.
5. **Validate and package it.** An authorized publish operator or setup administrator chooses **Package for playback**. Packaging must remain private.
6. **Verify privacy before approval.** The recording must not appear or play in the resident portal yet.
7. **Approve Portal publication.** In **Publish**, select only **Portal** and approve it.
8. **Confirm resident playback.** Open the resident portal in a second browser and play the approved recording. Unapproved media must remain private.

Common Operator Questions

What happens if Wi-Fi drops mid-meeting? The stock build does not claim automatic source-drop detection or slate failover (automatically switching to a hold slide when the video source drops) without a

verified media probe and station-specific egress proof (verified evidence that the station’s outbound stream setup actually works). Follow the station’s tested fallback procedure. Never assume the local recording continued: confirm the recording and finalization status before publishing.

Can I trim the start and end of the recording? The Assets editor is a scrub-and-metadata editor: it can save trim points and chapters, and it keeps the original. It does not claim automatic public re-rendering. Package the asset again after changing trim metadata, then review and publish it.

Where do captions come from? CivicCast generates captions automatically from the meeting audio using a local AI model — your audio never leaves the station unless you turn on a cloud model in settings. A records clerk reviews and corrects the captions before the recording is published.

How do residents find a recording? The public website has a **Browse** page with search and filters. Each recording also has a shareable link with a copy button. If your station tags recordings by meeting body (e.g. *City Council*, *Planning Commission*), residents can filter by that.

Do I need to upload anything to YouTube or Apple Podcasts myself? Optional providers require station credentials, configuration, and separate controlled proof. The clean-install walkthrough proves Portal-only publication; it does not claim that YouTube, Internet Archive, podcast directories, or other external providers are ready for a station.

How does a recording become public? In **Assets**, package the validated recording. Packaging prepares private HLS media but does not publish it. In **Publish**, approve the **Portal** surface. Only a successful Portal approval makes the recording metadata and HLS media resident-visible.

What does the restore rehearsal prove? The System Health action runs a real, isolated database restore drill. In the clean-host walkthrough it verified 95 tables and crash recovery. It does not prove recovery of media, configuration, or credentials; those require separate station backup and restore procedures.

How do I send support a bundle from Windows? In **System Health**, add a short note, choose **Create support bundle**, then choose **Download support bundle**. Send the downloaded JSON file and its displayed SHA-256 through the private beta support channel.

What if there’s an emergency during a meeting? CivicCast contains optional EAS (Emergency Alert System) / CAP (Common Alerting Protocol) software surfaces, but CivicCast does not claim a hardware-certified EAS relay or station take-over path. Use the station’s existing certified emergency procedure unless that integration has separate field proof. If EAS is not configured and proven, CivicCast does not interrupt the broadcast.

The meeting is closed-door — can I still record it without broadcasting? Yes. On **Run Meeting**, choose **Record only** instead of **Live**. The recording is stored privately until staff publish it.

When to Ask for Help

Send this to your IT lead, station admin, or open a GitHub issue at the project’s repository:

- The “Run Meeting” page says **Do not broadcast yet** and the yellow notice points to a step you do not have access to.
- The live stream shows on the operator console but residents see a blank or “stream offline” page on the public website.

- Captions are blank, garbled, or stuck on one line.
- A recording is missing from the **Assets** list after a meeting.
- CivicCast asks for a recovery code and the kit cannot be found.

For routine workflow questions, use the role-specific guides:

Admin Quick Guide

- [Admin Guide](#) — first install, recovery, backups.

Meeting Operator Quick Guide

- [Meeting Operator Guide](#) — night-of-meeting.

Records Clerk Quick Guide

- [Records Clerk Guide](#) — caption review, publish.
- [Operator Language Guide](#) — the words the console uses and what they mean.

For a problem the role guides do not cover, the project's GitHub Issues page is the right place. The Windows installer, operator console, and resident portal each include a **Report a beta issue** link that opens the project's bug template. Do not paste passwords, recovery codes, staff tokens, subscriber lists, private meeting material, or unedited meeting recordings into a bug report.

Section B — Technical Reference

This section is for the IT staff, integrators, and operators who maintain the station. It assumes a working knowledge of the shell, environment variables, and HTTP services.

For deeper command, credential, certificate, storage, model, and release-proof detail, see [Technical Operations Reference](#). For the narrative install procedure on Windows, see [INSTALL-WINDOWS.md](#).

Install And First Boot

The source supports two installation paths. Windows beta use is limited to the bounded recorded-media workflow while each new signed candidate repeats the genuine clean-host run first completed by rc14:

1. **Windows installer (controlled beta for testing).** The release .exe from the GitHub Release page registers a Windows service through the SCM, which supervises the control plane, Postgres, NATS, and the media workers from a bundled runtime under C:\Program Files\CivicCast (Native)\, and walks the operator through a recovery-kit and first-admin flow. Verify the SHA-256 hash against the release's .sidecar.json before running it. Do not assume the candidate is signed: compare its actual Authenticode status and publisher with the exact approved handoff. See [INSTALL-WINDOWS.md](#).

Leave at least **5 GB free** for the base installation. Recordings, station media, backups, and downloaded caption models require additional storage.

Local AI models (rc17 and later). The local AI models (Ollama summary and translation models, roughly 15-20 GB combined) are large; CivicCast ensures the same three-tag target set and downloads only the tags still missing, automatically in the background after the base install finishes, not before,

and a slow or failed model download does not block the operator console from opening.

2. Source / uv (for developers and integrators).

```
uv sync --all-extras --group dev
uv run civiccast --version
uv run civiccast doctor
uv run uvicorn civiccast.app:app --reload
```

Without DATABASE_URL, CivicCast starts in local setup mode and prepares installer-managed durable SQLite storage. Set DATABASE_URL to use Postgres in production.

After install, open the operator console at <http://localhost:8000/operator/> (or the installer-provided operator handoff URL), confirm **System Health** is green, save the recovery kit, and run a private first-broadcast rehearsal before the first public meeting.

Roles And Permissions

CivicCast has exactly five roles, defined in `civiccast/auth/roles.py`. Every operator-facing endpoint requires one of them.

Role	Purpose	Owns
setup_admin	First-install identity. Configures the station, recovery, storage, credentials, providers, and channel settings.	Setup wizard, channel automation, credentials, provider toggles, recovery codes.
meeting_operator	The person running tonight's meeting.	Run Meeting, preflight, start/stop broadcast, live captions tap.
records_clerk	Reviews and publishes the public record.	Caption review, summary review, signed-record export, publish approvals, chapter editing.
publish_operator	Day-to-day program-guide and cable-channel staff.	Program guide, schedule, bulletins, underwriting trafficking, asset metadata.
support_admin	Diagnostics and support. Can collect a redacted bundle and see system health, but cannot change records or publish.	Support bundles, system-health detail, logs.

Role aliases (kebab-case and dotted) are accepted by the API but canonical-cased forms are the ones written to the database. A single staff member can hold multiple roles.

Staff bearer tokens remain valid until revoked or rotated. Lifecycle-token secrets are shown once; CivicCast stores a salted PBKDF2 verifier plus a full-secret SHA-256 fingerprint used only for exact admission after a peer's failure budget is saturated. Legacy environment tokens remain valid while their configuration remains active only when they use the current versioned format from `civiccast token generate-env`. Headless callers (CI, the egress worker, cable packaging scripts) authenticate via `CIVICCAST_STAFF_TOKENS` or the per-token store managed through `civiccast token ... subcommands`.

Environment Variables

The exhaustive list of `CIVICCAST_*` variables CivicCast reads is below. Defaults are documented in the source comment of the consuming module. None of these are required for a default Windows installer flow — the installer writes a configuration file with the defaults that a typical station needs.

Identity, storage, and first-run

- **CIVICCAST_STATION_ID** — Short station identifier used in evidence files and headend handoff.
- **CIVICCAST_STATION_TZ** — IANA timezone (e.g. `America/New_York`) for schedules and as-run reports. Optional override: normally the service reads the timezone the operator chose during first-admin setup (persisted in the local station-state file); set this only to force a different zone without re-running setup.
- **CIVICCAST_ROOT** — Override for the durable-data root. Default: the installer-managed path.
- **CIVICCAST_CONFIG_DIR** — Config-file root. Default: under `CIVICCAST_ROOT`.
- **CIVICCAST_MANAGED_STORAGE_DIR** — Where SQLite durable storage lives when no `DATABASE_URL` is set.
- **CIVICCAST_BACKUP_DIR** — Managed backup target used by installer backup verification and disaster-recovery workflows.
- **CIVICCAST_UPLOAD_DIR** — Operator and contributor upload landing.
- **CIVICCAST_UPLOAD_MAX_BYTES** — Per-upload cap.
- **CIVICCAST_EVIDENCE_DIR** — Where proof artifacts (continuity proof, soak logs) are written.
- **CIVICCAST_VERSION** — Reported and update-checked version pins.
- **CIVICCAST_AVAILABLE_VERSION** — Reported and update-checked version pins.
- **CIVICCAST_SETUP_NONCE** — First-run protection for the setup wizard.
- **CIVICCAST_REQUIRE_SETUP_NONCE** — First-run protection for the setup wizard.
- **CIVICCAST_ALLOW_FIRST_ADMIN_RESET** — Permit a `setup_admin` reset after first-admin loss.
- **CIVICCAST_ALLOW_EPHEMERAL_STORES** — Permit in-memory stores. **Tests only.**
- **CIVICCAST_ALLOW_INSECURE_MANIFEST** — Permit an unsigned install manifest. **Tests only.**

Staff identity and tokens

- **CIVICCAST_STAFF_TOKENS** — Semicolon-separated `token:operator_id:display_name:role[,role...]` entries for explicit headless or recovery compatibility. Roles are required; generate each versioned random bearer secret with `civccast token generate-env`. Prefer lifecycle tokens for routine operation.
- **CIVICCAST_STAFF_TOKENS_FALLBACK_WITH_DB** — Allow the env-var tokens to coexist with DB-issued ones.
- **CIVICCAST_ALLOW_DETERMINISTIC_STAFF_TOKEN** — Use deterministic or short fixture tokens. **Tests only; never enable at a station.**
- **CIVICCAST_AUTH_ACK** — Acknowledgement gate for the auth subsystem on first run.
- **CIVICCAST_AUTH_RATE_LIMIT** — Failed-authentication budget. Default: 10. Setup uses peer-plus-route keys; staff bearer auth uses one observed-peer key across all staff routes. Valid staff tokens do not count.

- **CIVICCAST_AUTH_RATE_LIMIT_WINDOW_SECONDS** — Sliding failure-window duration in seconds. Default: 60.
- **CIVICCAST_REAL_BOUNDARY_TOKEN_FILE** — File holding the “real provider” enable token (prevents accidental live calls).

Channel egress, GStreamer engine, automation

- **CIVICCAST_EGRESS_ENGINE** — gstreamer (default, S15) or ffmpeg-concat (legacy fallback).
- **CIVICCAST_EGRESS_WORK_DIR** — Per-channel temporary directory for egress plans.
- **CIVICCAST_EGRESS_EMBED_CAPTIONS** — Attempts the native GStreamer caption-SEI path when the required GStreamer elements are available. The Windows installer stages the CivicCast-bundled private GStreamer runtime under the install root’s runtime\dependencies\gstreamer and verifies cccombiner, ccconverter, h264ccinserter, and tttocea608 with the bundled gst-inspect-1.0.exe; if any required element remains unavailable, setup stops with an explicit native caption-SEI runtime error instead of claiming the embed path is ready.
- **CIVICCAST_GST_ALLOW_HARDWARE_DECODE** — Optional expert override. By default, the GStreamer worker demotes GPU H.264/H.265 decoders so live UDP/SRT decode stays in system memory before the CPU conform/encode chain. Set to 1 only when the station has validated its hardware decode path end-to-end.
- **CIVICCAST_CHANNEL_AUTOMATION** — Enable the channel automation driver and its poll cadence.
- **CIVICCAST_CHANNEL_AUTOMATION_POLL_SECONDS** — Enable the channel automation driver and its poll cadence.
- **CIVICCAST_AUTOSCHEDULE** — Enable the S19 query-driven autoscheduler worker.
- **CIVICCAST_PROGRAM_LOG_WORKER** — Program-log materializer cadence.
- **CIVICCAST_PROGRAM_LOG_POLL_SECONDS** — Program-log materializer cadence.
- **CIVICCAST_PROGRAM_LOG_HORIZON_HOURS** — Program-log materializer cadence.
- **CIVICCAST_FINALIZATION_WORKER** — Recording-to-asset finalizer (S7) controls.
- **CIVICCAST_FINALIZATION_POLL_SECONDS** — Recording-to-asset finalizer (S7) controls.
- **CIVICCAST_FINALIZATION_MAX_ATTEMPTS** — Recording-to-asset finalizer (S7) controls.
- **CIVICCAST_FINALIZATION_BACKOFF_SECONDS** — Recording-to-asset finalizer (S7) controls.
- **CIVICCAST_FINALIZATION_SETTLE_SECONDS** — Recording-to-asset finalizer (S7) controls.
- **CIVICCAST_FINALIZATION_NEVER_APPEARED_SECONDS** — Recording-to-asset finalizer (S7) controls.
- **CIVICCAST_FINALIZATION_RUNNING_LEASE_SECONDS** — Recording-to-asset finalizer (S7) controls.
- **CIVICCAST_RETENTION_WORKER** — Retention worker controls.
- **CIVICCAST_RETENTION_POLL_SECONDS** — Retention worker controls.
- **CIVICCAST_BULLETIN_EXPIRY** — CG bulletin expiry worker.
- **CIVICCAST_RELOAD_TIMEOUT_S** — Worker-supervision timeouts.
- **CIVICCAST_STALL_TIMEOUT_S** — Worker-supervision timeouts.

Captions, EAS, alerting

- **CIVICCAST_CAPTION_TAP** — Live caption tap configuration.

- **CIVICCAST_CAPTION_TAP_DIR** — Live caption tap configuration.
- **CIVICCAST_CAPTION_TAP_POLL_SECONDS** — Live caption tap configuration.
- **CIVICCAST_CAPTION_TAP_SEGMENT_SECONDS** — Live caption tap configuration.
- **CIVICCAST_CAPTION_FEED_POLL_SECONDS** — Caption feed and decode-back proof cadence.
- **CIVICCAST_CAPTION_PROOF_POLL_SECONDS** — Caption feed and decode-back proof cadence.
- **CIVICCAST_EAS** — EAS subsystem on/off and surface cadence.
- **CIVICCAST_EAS_AUTO_SURFACE** — EAS subsystem on/off and surface cadence.
- **CIVICCAST_EAS_POLL_SECONDS** — EAS subsystem on/off and surface cadence.
- **CIVICCAST_ALERTING** — Master switch for the operational alerting service.
- **CIVICCAST_ALERT_CREDENTIALS_FILE** — JSON file holding Twilio SMS credentials.
- **CIVICCAST_EVENTS** — Event-bus on/off.

CDN, public base URL, trusted proxies

- **CIVICCAST_PUBLIC_BASE_URL** — The URL residents see (<https://broadcast.town.gov>).
- **CIVICCAST_LIVE_MANIFEST_BASE_URL** — Override for the live-stream manifest base.
- **CIVICCAST_CDN_PROVIDER** — bunny, cloudflare, or cloudflare_r2. Wires the trusted-proxy resolver to the provider's edge CIDRs.
- **CIVICCAST_CDN_STUB_ROOT** — Local stub directory for CDN dry-run testing.
- **CIVICCAST_TRUSTED_PROXY_CIDRS** — Explicit comma-separated CIDR list, unioned with the provider preset.
- **CIVICCAST_TRUST_PRIVATE_PROXIES** — Default false. When false, an X-Forwarded-For header is honored only from a proxy in **CIVICCAST_TRUSTED_PROXY_CIDRS**, so a direct caller cannot spoof its source address. Set true only when the station genuinely sits behind a trusted RFC1918/loopback reverse proxy that sets the header itself.
- **CIVICCAST_BUNNY_STORAGE_ZONE** — BunnyCDN credentials and hostname.
- **CIVICCAST_BUNNY_ACCESS_KEY** — BunnyCDN credentials and hostname.
- **CIVICCAST_BUNNY_CDN_HOSTNAME** — BunnyCDN credentials and hostname.
- **CIVICCAST_R2_ACCOUNT_ID** — Cloudflare R2 origin credentials.
- **CIVICCAST_R2_ACCESS_KEY_ID** — Cloudflare R2 origin credentials.
- **CIVICCAST_R2_SECRET_ACCESS_KEY** — Cloudflare R2 origin credentials.
- **CIVICCAST_R2_BUCKET** — Cloudflare R2 origin credentials.
- **CIVICCAST_R2_PUBLIC_BASE_URL** — Cloudflare R2 origin credentials.

TSA / records / RFC 3161

- **CIVICCAST_TSA_ENABLE** — Enable RFC 3161 timestamp signing on signed records.
- **CIVICCAST_TSA_URL** — TSA endpoint URL. Default: FreeTSA (<https://freetlsa.org/tsr>).
- **CIVICCAST_TSA_POLICY_OID** — Optional policy OID to assert in the TSA request.
- **CIVICCAST_VERAPDF_ARTIFACT_DIR** — veraPDF artifact directory for the signed-record export proof.

Providers (publish, archive, syndicate) These are set per kind. The default for every kind is the deterministic mock; set the per-kind variable to `real` and supply credentials to use the real adapter.

- **CIVICCAST_PROVIDER_INTERNET_ARCHIVE** — mock (default) or real.
- **CIVICCAST_PROVIDER_YOUTUBE** — mock (default) or real.
- **CIVICCAST_PROVIDER_MAIL** — mock (default) or real.
- **CIVICCAST_PROVIDER_WEBHOOK** — mock (default) or real.
- **CIVICCAST_PROVIDER_LOCAL_NAS** — mock (default) or real.
- **CIVICCAST_PROVIDER_API_KEY** — Provider credential bundle.
- **CIVICCAST_PROVIDER_CREDENTIALS_FILE** — Provider credential bundle.
- **CIVICCAST_PROVIDER_PROOFS_FILE** — Provider credential bundle.
- **CIVICCAST_IA_ACCESS_KEY** — Internet Archive S3-like credentials.
- **CIVICCAST_IA_SECRET_KEY** — Internet Archive S3-like credentials.
- **CIVICCAST_IA_COLLECTION** — Internet Archive S3-like credentials.
- **CIVICCAST_IA_ITEM_PREFIX** — Internet Archive S3-like credentials.
- **CIVICCAST_IA_ACCESS_VALUE** — Internet Archive S3-like credentials.
- **CIVICCAST_INTERNET_ARCHIVE_ACCESS_KEY** — Internet Archive S3-like credentials.
- **CIVICCAST_YOUTUBE_CLIENT_ID** — YouTube OAuth credentials and upload defaults.
- **CIVICCAST_YOUTUBE_CLIENT_SECRET** — YouTube OAuth credentials and upload defaults.
- **CIVICCAST_YOUTUBE_REFRESH_TOKEN** — YouTube OAuth credentials and upload defaults.
- **CIVICCAST_YOUTUBE_REFRESH_VALUE** — YouTube OAuth credentials and upload defaults.
- **CIVICCAST_YOUTUBE_PRIVACY** — YouTube OAuth credentials and upload defaults.
- **CIVICCAST_YOUTUBE_MEDIA_ROOT** — YouTube OAuth credentials and upload defaults.
- **CIVICCAST_SMTP_HOST** — SMTP for the email subscriber adapter.
- **CIVICCAST_SMTP_PORT** — SMTP for the email subscriber adapter.
- **CIVICCAST_SMTP_USERNAME** — SMTP for the email subscriber adapter.
- **CIVICCAST_SMTP_PASSWORD** — SMTP for the email subscriber adapter.
- **CIVICCAST_SMTP_STARTTLS** — SMTP for the email subscriber adapter.
- **CIVICCAST_SMTP_FROM** — SMTP for the email subscriber adapter.
- **CIVICCAST_EMAIL_FROM** — SMTP for the email subscriber adapter.
- **CIVICCAST_EMAIL_SMTP_URL** — SMTP for the email subscriber adapter.
- **CIVICCAST_NAS_TARGET** — Local NAS archive target.
- **CIVICCAST_NAS_ARCHIVE_PATH** — Local NAS archive target.

Subscribe (email + webhook)

- **CIVICCAST_SUBSCRIBE_TOKEN_SECRET**
HMAC secret for magic-link confirmation tokens.
- **CIVICCAST_SUBSCRIBE_ENCRYPTION_KEY**
Encryption key for subscriber records at rest.
- **CIVICCAST_SUBSCRIBE_LEGACY_ENCRYPTION_KEYS**
Rotation helpers for older secrets.
- **CIVICCAST_SUBSCRIBE_LEGACY_TOKEN_SECRETS**
Rotation helpers for older secrets.
- **CIVICCAST_SUBSCRIBE_ACCEPT_V08_LEGACY_SECRETS**
Rotation helpers for older secrets.

- **CIVICCAST_SUBSCRIBE_SECRETS_FILE**
File-backed alternative to the env-var secret.
- **CIVICCAST_SUBSCRIBE_RATE_LIMIT**
Subscribe form rate limiting.
- **CIVICCAST_SUBSCRIBE_RATE_LIMIT_WINDOW_SECONDS**
Subscribe form rate limiting.
- **CIVICCAST_SUBSCRIBER_WEBHOOK_SECRET**
Webhook signature secret.
- **CIVICCAST_WEBHOOK_SIGNING_VALUE**
Webhook signature secret.
- **CIVICCAST_WEBHOOK_RETRY_WORKER**
Webhook retry worker controls.
- **CIVICCAST_WEBHOOK_RETRY_BACKOFF_SECONDS**
Webhook retry worker controls.
- **CIVICCAST_WEBHOOK_RETRY_MAX_ATTEMPTS**
Webhook retry worker controls.
- **CIVICCAST_WEBHOOK_RETRY_POLL_SECONDS**
Webhook retry worker controls.
- **CIVICCAST_WEBHOOK_TIMEOUT_SECONDS**
Webhook retry worker controls.

CIVICCAST_ALLOW_DETERMINISTIC_SUBSCRIBE_SECRETS enables deterministic test secrets. It is for tests only and must never be enabled at a station.

Analytics

- **CIVICCAST_PUBLIC_ANALYTICS_KEY** — Public key the resident-portal beacon uses.
- **CIVICCAST_PUBLIC_ANALYTICS_ALLOWED_ORIGINS** — Analytics ingestion controls.
- **CIVICCAST_PUBLIC_ANALYTICS_RATE_LIMIT_PER_MINUTE** — Analytics ingestion controls.
- **CIVICCAST_TRUSTED_ANALYTICS_RATE_LIMIT_PER_MINUTE** — Analytics ingestion controls.
- **CIVICCAST_ANALYTICS_RATE_LIMIT_MAX_BUCKETS** — Analytics ingestion controls.
- **CIVICCAST_ANALYTICS_RETENTION_DAYS** — Days to keep raw analytics events.
- **CIVICCAST_ANALYTICS_TRUSTED_PROXY_CIDRS** — Analytics-specific trusted-proxy CIDRs.

ActivityPub federation

- **CIVICCAST_ACTIVITYPUB_MODE** — off, lab, or live.
- **CIVICCAST_ACTIVITYPUB_BASE_URL** — Identity.
- **CIVICCAST_ACTIVITYPUB_HANDLE** — Identity.
- **CIVICCAST_ACTIVITYPUB_DISPLAY_NAME** — Identity.
- **CIVICCAST_ACTIVITYPUB_PRIVATE_KEY_PATH** — HTTP-signature keypair.
- **CIVICCAST_ACTIVITYPUB_PUBLIC_KEY_PEM** — HTTP-signature keypair.
- **CIVICCAST_ACTIVITYPUB_ALLOWLIST** — Federation domain filters.
- **CIVICCAST_ACTIVITYPUB_ALLOWLIST_FILE** — Federation domain filters.

- **CIVICCAST_ACTIVITYPUB_BLOCKLIST** — Federation domain filters.
- **CIVICCAST_ACTIVITYPUB_BLOCKLIST_FILE** — Federation domain filters.
- **CIVICCAST_ACTIVITYPUB_AUTHORIZED_FETCH** — Strict fetch mode and lab-loopback override.
- **CIVICCAST_ACTIVITYPUB_LAB_ALLOW_LOCAL** — Strict fetch mode and lab-loopback override.
- **CIVICCAST_ACTIVITYPUB_INBOX_RATE_LIMIT** — Retry and rate-limit controls.
- **CIVICCAST_ACTIVITYPUB_INBOX_RATE_WINDOW_SECONDS** — Retry and rate-limit controls.
- **CIVICCAST_ACTIVITYPUB_RETRY_WORKER** — Retry and rate-limit controls.
- **CIVICCAST_ACTIVITYPUB_RETRY_BACKOFF_SECONDS** — Retry and rate-limit controls.
- **CIVICCAST_ACTIVITYPUB_RETRY_MAX_ATTEMPTS** — Retry and rate-limit controls.
- **CIVICCAST_ACTIVITYPUB_RETRY_POLL_SECONDS** — Retry and rate-limit controls.

Headend handoff: NDI, SDI, TSDuck

- **CIVICCAST_NDI_RELAY** — NDI output relay configuration.
- **CIVICCAST_NDI_SENDER** — NDI output relay configuration.
- **CIVICCAST_NDI_FFMPEG** — NDI output relay configuration.
- **CIVICCAST_NDI_RUNTIME_DIR** — NDI output relay configuration.
- **CIVICCAST_SDI_RELAY** — SDI output relay (descope from 3.0 default but available).
- **CIVICCAST_SDI_FFMPEG** — SDI output relay (descope from 3.0 default but available).
- **CIVICCAST_TSDUCK_HOME** — TSDuck compliance-probe toolchain. CivicCast fetches and installs a pinned, checksum-verified TSDuck portable build into a contained per-user directory on demand (no admin rights, no system installer) and verifies `tsp --version` before setup can claim the runtime is ready.
- **CIVICCAST_TSDUCK_PATH** — TSDuck compliance-probe toolchain. CivicCast fetches and installs a pinned, checksum-verified TSDuck portable build into a contained per-user directory on demand (no admin rights, no system installer) and verifies `tsp --version` before setup can claim the runtime is ready.
- **CIVICCAST_TSDUCK_RCVBUF_BYTES** — TSDuck compliance-probe toolchain. CivicCast fetches and installs a pinned, checksum-verified TSDuck portable build into a contained per-user directory on demand (no admin rights, no system installer) and verifies `tsp --version` before setup can claim the runtime is ready.
- **CIVICCAST_TSDUCK_NETWORK_TESTS** — TSDuck compliance-probe toolchain. CivicCast fetches and installs a pinned, checksum-verified TSDuck portable build into a contained per-user directory on demand (no admin rights, no system installer) and verifies `tsp --version` before setup can claim the runtime is ready.
- **CIVICCAST_CABLE_PACKAGE_OUTPUT_DIR** — Cable file-package output.
- **CIVICCAST_CABLE_CAPTIONS_DIR** — Cable file-package output.

Remote contribution & production control room

- **CIVICCAST_REMOTE_CONTRIBUTION**
S17 VDO.Ninja guest-input service.
- **CIVICCAST_REMOTE_CONTRIBUTION_HOME**
S17 VDO.Ninja guest-input service.

- **CIVICCAST_REMOTE_CONTRIBUTION_POLL**
S17 VDO.Ninja guest-input service.
- **CIVICCAST_REMOTE_CONTRIBUTION_VDO_URL**
S17 VDO.Ninja guest-input service.
- **CIVICCAST_CONTRIBUTOR_STORE_PATH**
Contributor record store and upload landing.
- **CIVICCAST_CONTRIBUTOR_UPLOAD_DIR**
Contributor record store and upload landing.
- **CIVICCAST_TURN_HOST / CIVICCAST_TURN_PORT**
The TURN server guests connect through to punch out from behind NAT. coturn has no native Windows build, so on Windows CivicCast does **not** run a local coturn process: point these two variables at a **documented external TURN server** — a coturn instance on a separate Linux/BSD host, or a managed TURN provider — and leave CIVICCAST_COTURN_COMMAND unset. The operator console’s Remote Contribution screen (Diagnostics drawer, support_admin role) shows the currently-configured host/port, a **Test TURN connectivity** button that probes it right now, and the same guidance text under “How to point this station at coturn.” On Linux/macOS these still work the same way if you’d rather point at an external server than run coturn locally.
- **CIVICCAST_COTURN_COMMAND**
Linux/macOS only — the local coturn launch command, if you’re running coturn on the station itself rather than pointing at an external server (see CIVICCAST_TURN_HOST above). Leave unset on Windows; there is no native Windows build.
- **CIVICCAST_VDO_COMMAND**
TURN/VDO co-process commands.
- **CIVICCAST_CONTROL_ROOM_TSR_URL**
S16 production-control-room TSR endpoint.
- **CIVICCAST_TSR_PORT**
S16 production-control-room TSR endpoint.
- **CIVICCAST_TSR_NO_LISTEN**
S16 production-control-room TSR endpoint.
- **CIVICCAST_PORTAL_BASE**
Resident portal URL handles.
- **CIVICCAST_PORTAL_TOKEN**
Resident portal URL handles.
- **CIVICCAST_RESIDENT_PORTAL_URL**
Resident portal URL handles.

AI model dispatch

- **CIVICCAST_OLLAMA_BASE_URL** — Local Ollama endpoint for the on-station model tier.
- **CIVICCAST_PROVIDER_API_KEY** — Cloud-provider key for Ollama Cloud or OpenRouter. (Prefer the keyring path: `civiccast model set-provider-key`.)

Platform broker (event bus)

- **CIVICCAST_BROKER_MODE** — process (default) or nats.
- **CIVICCAST_NATS_URL** — NATS connection.
- **CIVICCAST_NATS_STREAM** — NATS connection.
- **CIVICCAST_NATS_DURABLE** — NATS connection.
- **CIVICCAST_NATS_CA_FILE** — NATS connection.
- **CIVICCAST_NATS_CLIENT_CERT_FILE** — NATS connection.
- **CIVICCAST_NATS_CLIENT_KEY_FILE** — NATS connection.

Diagnostics and overrides

- **CIVICCAST_OPERATOR_CONSOLE_DIST** — Override the prebuilt frontend bundles.
- **CIVICCAST_OPERATOR_CONSOLE_URL** — Override the prebuilt frontend bundles.
- **CIVICCAST_PUBLIC_PORTAL_DIST** — Override the prebuilt frontend bundles.
- **CIVICCAST_APP_BUILD_ARTIFACTS_DIR** — OTT app-build orchestrator paths.
- **CIVICCAST_APP_BUILD_STORE_PATH** — OTT app-build orchestrator paths.
- **CIVICCAST_APP_PLATFORM_CONFIG_PATH** — OTT app-build orchestrator paths.
- **CIVICCAST_CERT_ROOT** — Local-CA service-cert root override.
- **CIVICCAST_REQUIRE_FCC_ACK** — Require an FCC-rule acknowledgement before EAS surfaces.
- **CIVICCAST_ROLLBACK_ARTIFACT_PATH** — Path to a rollback artifact.
- **CIVICCAST_TESTER_AVAILABLE_VERSION** — Tester-rig overrides; rarely used in production.
- **CIVICCAST_TESTER_OPS_STATE_PATH** — Tester-rig overrides; rarely used in production.
- **CIVICCAST_PLAYBACK_POLICY_STATE_PATH** — Tester-rig overrides; rarely used in production.
- **CIVICCAST_STATION_STATE_PATH** — Tester-rig overrides; rarely used in production.
- **CIVICCAST_SUPPORT_BUNDLE_DIR** — Tester-rig overrides; rarely used in production.
- **CIVICCAST_RUN_POSTGRES_TESTS** — Run Postgres-only test suites against an external server.

Credential Store

Three families of credentials are kept in a dedicated store rather than environment variables so a station can rotate them without restarting the service.

Subsystem	Credential	Stored As	Configuration
Alerting (Twilio SMS)	account SID, auth token, sender number	JSON file referenced by CIVICCAST_ALERT_CREDENTIALS_FILE , with file-mode enforcement on POSIX hosts	Set the path env var, write the JSON, restart.
Subscription paywall (Stripe)	publishable key, secret key, webhook signing secret	Paywall service secret store; never written to logs	Configured in the operator console under <i>Settings</i> → <i>Paywall</i> . Default off (S26, V1.x).

Subsystem	Credential	Stored As	Configuration
RFC 3161 TSA	endpoint URL, optional policy OID	CIVICCAST_TSA_URL and CIVIC- CAST_TSA_POLICY_OID	FreeTSA is the default; bring your own TSA in production.
Cloud AI providers	per-provider API key	OS keyring, via civiccast model set-provider-key	Keys are write-only — never echoed back.

The store never logs a credential. The most an operator sees is a *stored* / *not-stored* line and a credential reference label.

CLI Reference

civiccast is a Typer-based CLI. Most operators rarely need it — the operator console is the day-to-day surface — but integrators and on-call use it heavily.

```
civiccast --version
```

```
civiccast doctor [--json] [--disk PATH]
```

```
civiccast installer plan \
```

```
    [--profile public-meetings] [--recommended-tier tier-1]
```

```
civiccast installer health-check [--profile public-meetings]
```

```
civiccast installer platform-plan [--os-family linux|macos]
```

```
    # Generic Linux/macOS bootstrap planning only. Windows deployment
```

```
    # readiness is decided separately by the native station's own
```

```
    # activation state (civiccast.installer.service), not this plan.
```

```
civiccast installer verify-package --artifact PATH --sidecar PATH
```

```
civiccast installer summary
```

```
civiccast installer beta-handoff \
```

```
    [--release-manifest PATH] [--clean-windows-evidence PATH]
```

```
civiccast model download [--offline-bundle --bundle-dir PATH] \
```

```
    [--profile NAME] [--dry-run]
```

```
civiccast model state
```

```
civiccast model import-offline --bundle-dir PATH \
```

```
    --expected file=sha256 [--expected ...]
```

```
civiccast model set-provider-key ollama-cloud|openrouter \
```

```
    [--key SECRET] [--clear]
```

```
civiccast cert rotate IDENTITY
```

```
civiccast token ... (managed by the operator console; see /api/tokens)
```

```
civiccast cable package --asset-id ID --title T --media F \
```

```
    --captions F --output-dir D
```

```
civiccast cable ndi-check
```

```
civiccast cable ndi-plan --media F --ndi-name N \
```

```
    [--muxer libndi_newtek]
```


`civiccast activitypub ...` (key + config helpers; see `civiccast/activitypub/`)

`civiccast egress run --channel-id CH [--work-dir D] \`

`[--poll-seconds 2] [--once]`

`civiccast egress verify --channel-id CH [--seconds 10]`

`civiccast egress recovery-proof --channel-id CH \`

`--measured-seconds N ...`

`civiccast egress continuity-proof --source-plan-json J \`

`--config-json J \`

`--output-path F`

`civiccast egress srt-continuity-proof \`

`--source-plan-json J --config-json J \`

`--sender-url U --receiver-url U \`

`--receiver-output-path F`

`civiccast egress caption-decode-proof --channel-id CH ...`

`civiccast egress trim-health [--older-than-days 30] [--dry-run]`

`civiccast live-takeover ...` (cut a channel to a live source and return it)

Every subcommand supports `--json` for machine-readable output. Long-running ones (`civiccast egress run`) respond to SIGTERM and SIGINT cleanly.

Headless callers should authenticate via `CIVICCAST_STAFF_TOKENS` rather than passing tokens on the command line. Recovery codes, provider keys, and Stripe secrets must never appear on a command line in production.

Troubleshooting Matrix

Symptom	First Check	Then
Operator console shows Do not broadcast yet .	Open System Health. Read each not-ready row.	The line points to the next action (Setup → Storage, Setup → Credentials, etc.).
Live stream visible on operator console, blank on the public website.	<code>CIVICCAST_PUBLIC_BASE_URL</code> matches what residents actually load.	Confirm <code>CIVICCAST_CDN_PROVIDER</code> and <code>CIVICCAST_TRUSTED_PROXY_CIDRS</code> if the station is behind BunnyCDN / Cloudflare.
Captions empty or stuck.	<code>civiccast doctor</code> — confirm CPU/RAM tier; <code>CIVICCAST_OLLAMA_BASE_URL</code> reachable.	Look in the caption-tap directory (<code>CIVICCAST_CAPTION_TAP_DIR</code>) for stale segments.
Recording missing after a meeting.	<code>civiccast egress run --once --channel-id CH</code> for a snapshot of the egress worker state.	Check the recording finalizer worker (<code>CIVICCAST_FINALIZATION_WORKER</code>) and its logs.

Symptom	First Check	Then
Twilio SMS alerts not firing.	<code>CIVICCAST_ALERTING=true</code> ; <code>CIVIC-CAST_ALERT_CREDENTIALS_FILE</code> points at a readable JSON.	Run an alert self-test from the operator console (System Health → Alerting).
EAS surface stuck on needs FCC ack .	Confirm the FCC acknowledgement is enabled and recorded in setup.	Re-open the EAS settings screen and confirm.
Trim re-render never finishes.	Recording finalizer worker is running.	Inspect the <code>civic-cast/live/finalization_worker</code> logs for the asset id.
Behind a CDN, source-IP filters block legitimate residents. <code>civiccast doctor</code> reports a tier below what the box should provide.	<code>CIVICCAST_CDN_PROVIDER</code> is set and the preset is current. <code>--disk</code> points at the durable-storage volume.	Add explicit CIDRs in <code>CIVIC-CAST_TRUSTED_PROXY_CIDRS</code> . Re-run with the explicit <code>--disk</code> argument.
Cloud AI key looks set but dispatch is offline.	<code>civiccast model set-provider-key PROVIDER</code> — confirm <i>stored</i> .	Re-check that the model the operator picked is supported by the saved provider.

For deeper trouble — TSDuck failures, GStreamer decode-back proofs, veraPDF validation, certificate rotation, mTLS — see [Technical Operations Reference](#).

The Windows installer provisions the native beta runtime directly on the host, under the bundled runtime tree at `C:\Program Files\CivicCast (Native)\runtime\`: Python, FFmpeg/FFprobe, the CivicCast-bundled private GStreamer runtime with the caption-SEI elements, and Faster Whisper. TSDuck for cable verification is fetched and verified on demand into a contained per-user directory when cable verification is enabled — no admin rights, no system installer. NDI runtime/SDK, DeckLink hardware/drivers, app-store provider accounts, and live station headend equipment remain operator/provider supplied.

Local AI provisioning (rc17 and later). CivicCast also provisions the local Ollama runtime for on-station AI (reusing a healthy existing install if one is already present and installing a pinned version only when Ollama is absent), then ensures the same three-tag target set of standard summary and translation models, downloading only the tags still missing, in the background after the operator console is already reachable; a model-download failure is reported honestly without blocking the rest of the install.

Cross-references

- [Admin Guide](#) — first install, recovery, backups.
- [Meeting Operator Guide](#) — night-of-meeting.
- [Records Clerk Guide](#) — caption review, publish.
- [Technical Operations Reference](#) — exact commands, certificate operations, release proofs.
- [Operator Language Guide](#) — UI vocabulary.
- [Channel Egress Operator And Tester Runbook](#).

- [Windows Release Trust And Verification.](#)

Section C — Architecture Reference

This section is for developers, integrators, and external auditors who need to understand the source architecture and its relationship to the CivicCast 3.0 station-in-a-box specification. Every claim below cross-references a spec section or a concrete module path.

The canonical specification is at [docs/spec/3.0/civiccast-3.0-station-in-a-box-MASTER.md](#). The repo-verified per-section status manifest is at [docs/spec/3.0/ROADMAP.status.yaml](#). The migration-chain reconciliation history is at [docs/spec/3.0/RECONCILIATION.md](#).

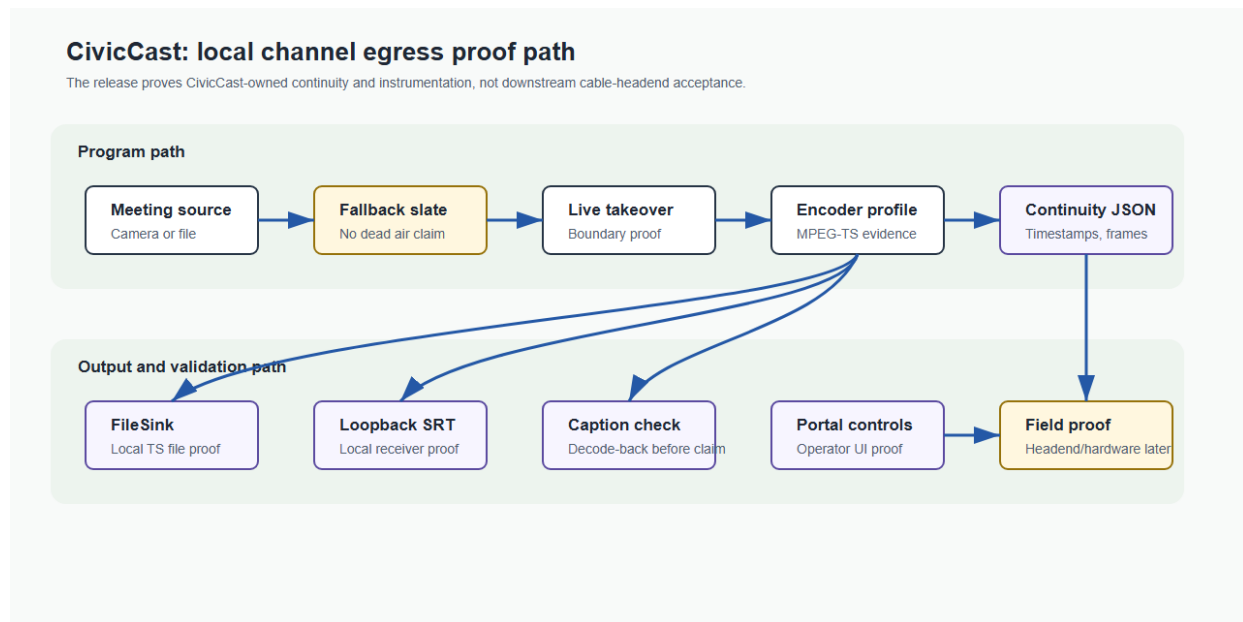


Figure 2: CivicCast channel-egress architecture

Subsystems And Module Layout

```

civiccast/
├─ app.py           FastAPI umbrella
├─ cli.py           Typer CLI (Section B)
├─ auth/           Identity, roles, staff tokens
├─ installer/      Commissioning wizard (S3), first-boot
├─ egress/         Channel egress; gst/ = S15 engine
│   └─ gst/engine.py  GstPlayayoutEngine
│   └─ service.py     Egress daemon + supervisor (S9)
├─ facility/       Commit-to-Air (S4) + force matrix (S5)
├─ cg/            CG bulletin board (S6)
├─ schedule/       Programlog, autoschedule (S7, S19)
├─ live/          Live meeting + finalization (S7 finalizer)
├─ recording/      Scheduled recording (S21)

```

— metadata/	Custom fields (S22)
— reporting/	As-run + EPG + franchise (S23)
— underwriting/	Spot management + affidavits (S24)
— agenda/	Meeting agenda + chapters (S25)
— paywall/	Subscription paywall (S26, default off)
— alerting/	Health + Twilio SMS (S8)
— eas/	Public-safety surface (S11)
— captions/	Live caption tap (S11)
— records/	Signed PDF + RFC 3161 (S7 + records lane)
— analytics/	Resident-portal beacons + audience reports (S14)
— ai_models/	Operator-chosen model dispatch (S13)
— app_platform/	OTT build orchestration (S12)
— apps/ott-native/	Native source for 6 OTT targets (S12 D1 closed)
— control_room/	TSR over OBS/vMix/ATEM (S16, optional)
— live/contribution/	VDO.Ninja remote guests (S17, optional)
— archive/	Internet Archive + local NAS
— syndicate/	YouTube
— subscribe/	Email + webhook subscribers
— activitypub/	ActivityPub federation
— stream/cdn/	BunnyCDN / Cloudflare R2 origin
— common/trusted_proxy.py	CDN-aware client-IP resolver

The Five-Role Model

Defined in [civicast/auth/roles.py](#):

```
KNOWN_ROLES = (
    "setup_admin",
    "meeting_operator",
    "records_clerk",
    "publish_operator",
    "support_admin",
)
```

The authoritative permission matrix is the per-endpoint dependency in each router (`require_role(...)`). A staff member can hold multiple roles; the OR of their roles is what the dependency checks. Roles are stable across releases; new roles require a migration and a deprecation window.

The Alembic Migration Chain

The migration chain is single-headed at `0072_normalize_recording_file_uris`. The shape — including the late-landing sibling for S21 — is below in ASCII form; the canonical version (with each revision's down-revision pointer and the reasoning for the sibling) is in [RECONCILIATION.md](#).

```
0049_per_sink_loudness
-> 0050_caption_proof_samples
-> 0051_public_safety_eas
-> 0052_secondary_audio
-> 0053_ai_model_configuration
```

```

-> 0054_custom_metadata_fields (S22)
-> 0055_asrun_and_epg (S23)
  |-- 0056_scheduled_recording (S21) --|
  `-- 0057_underwriting_spots (S24)    |
      -> 0058_meeting_agenda (S25)    |
      -> 0059_paywall_access (S26)    |
          0060_recording_paywall_merge
-> 0061_control_room_mode_gate
-> 0062_media_integrity_columns
-> 0063_producer_ops
-> 0064_control_room_health_and_versioning
-> 0065_recording_dropout_fields
-> 0066_hls_sink_kind
-> 0067_agenda_import_provenance
-> 0068_migrate_batches
-> 0069_control_room_session_surface_lock
-> 0070_grandfather_scheduled_to_published
-> 0071_published_blocks_overlap
-> 0072_normalize_recording_file_uris (HEAD)

```

0060 is the historical data-free merge revision that unified the 0056 sibling with the 0059 linear head. Later subsystem revisions extend that line through the current 0071 head. Automated schema-head tests enforce that single-head identity; [ROADMAP.status.yaml](#) tracks the capability evidence associated with the chain.

The S15 GStreamer Playout Engine

The S15 playout engine ([civicast/egress/gst/](#)) replaces the previous ffmpeg-relay model with a persistent GStreamer pipeline plus hot-swap. The two relevant pieces:

- `engine.py` — the `GstPlayoutEngine` itself. A persistent pipeline configured around `GstInterpipe` / input-selector for seamless source swap (S15, also master §10 step 0 + step 1).
- `graph.py` — pipeline graph definitions; the playout, file-sink, SRT, and NDI shapes used by the egress daemon.
- `worker.py` — the in-process worker the egress daemon supervises. This is what the beta release-artifact soak (step 13) targets.

gstreamer is the default engine, matching the v3.0 station-in-a-box specification’s recommendation and the native station bootstrap’s own runtime contract. Set `CIVICCAST_EGRESS_ENGINE=ffmpeg-concat` to opt back into the legacy ffmpeg-relay model for backwards compatibility.

The Three Protocol Seams

The three modules built last in the v3.0 sprint (scheduled recording, asset finalization, alerting) connect to the rest of the station via explicit typing.Protocol seams. The pattern lives in [civicast/recording/service.py](#), which is the most thorough example.

```

class CapturePipelineProtocol(Protocol):
    """Start, stop, and observe a capture pipeline for a scheduled recording."""

```

```

class AssetFinalizerProtocol(Protocol):
    """Hand a finished capture to the publish pipeline as a registered Asset."""

class AlertSinkProtocol(Protocol):
    """Receive scheduled-recording health / failure events as operator alerts."""

```

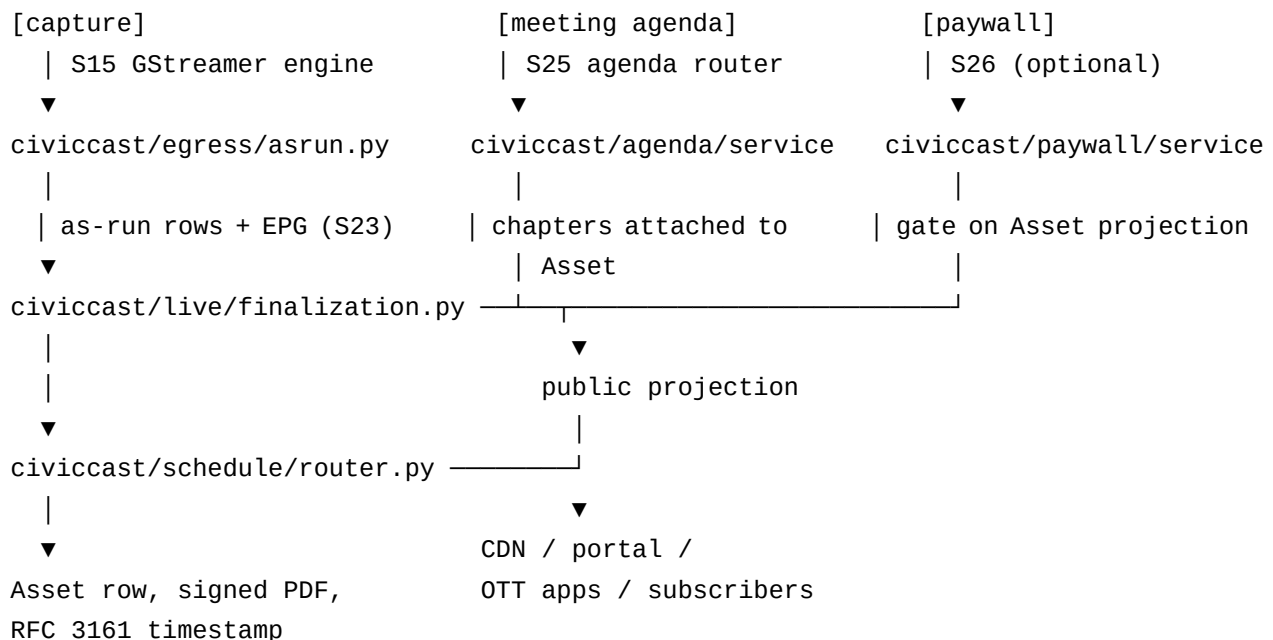
RecordingService.__init__ takes each Protocol as an optional parameter. When None, the feature is treated as disabled (no capture runs, no asset finalization, no alerts), which keeps the data layer honest. The production FastAPI factory now binds:

- CapturePipelineProtocol -> FfmpegScheduledCapturePipeline, which captures SDI/HDMI/NDI and network streams through the FFmpeg runtime.
- AssetFinalizerProtocol -> ScheduledRecordingAssetFinalizer, which probes the recorded file, validates ingest metadata, and registers a recorded Asset.
- AlertSinkProtocol -> RecordingAlertSink, which sends source and finalization failures to the S8 condition hub.

This shape is what lets S21 ship as a complete service + API + UI in its own slice without entangling the egress, capture, and alerting subsystems. Each Protocol is independently mockable and independently testable.

The Publish Pipeline

Recordings, meeting agendas, and chapter markers tie into the public Asset graph through the S7 media-lifecycle subsystem. The relevant flow:



Finalization and packaging create the private media package. Public routes also require published_at, which is set only after the Portal surface succeeds in the publish workflow. An unpublished package returns 404 from both public metadata and /media/vod; packaging is never treated as approval.

The schedule_adapter/asrun_recorder in civiccast/reporting/ record what actually went on air,

producing the as-run reports the franchise authority sees. EPG export (`civiccast/reporting/epg.py`) emits XMLTV for downstream guides.

The CDN-Aware Trusted-Proxy Resolver

Located at `civiccast/common/trusted_proxy.py`. Everything that consumes “real client IP” goes through this resolver: paywall rate limiter, recording cross-station guard, audit logger, analytics ingestion filter.

```

incoming request
  |
  ▼
request.client.host = X
  |
  ▼
walk X-Forwarded-For right-to-left
  |
  └─ if X is a trusted proxy → trust the chain
      trusted = CIVICCAST_TRUSTED_PROXY_CIDRS
                + provider preset (CIVICCAST_CDN_PROVIDER)
                + private/loopback (CIVICCAST_TRUST_PRIVATE_PROXIES)
  |
  └─ else → the immediate peer IS the real client
            (X-Forwarded-For is ignored – anti-spoof)

```

Provider presets ship for bunny and cloudflare / cloudflare_r2, snapshotted from the providers’ published edge-CIDR lists. Operators should reconcile the snapshot with the provider’s current documentation at deploy time; CivicCast does not auto-update at runtime.

Comparative Capability Status

The repository software-capability table is tracked in [ROADMAP.status.yaml](#). A shipped row means the software surface exists; it does not by itself prove the installer, station hardware, external provider, or field workflow:

PEG automation surface	CivicCast section	Status
Playout engine	S15	shipped (GStreamer)
Commit-to-Air / operator takeover controls	S4 / S5	shipped
CG / bulletin board	S6	shipped
Health + alerting	S8	shipped (Twilio SMS adapter is real)
OTT apps	S12	shipped (6 native source trees)
AI model selection	S13	shipped
Captions / Loudness / EAS	S11	shipped
Scheduling automation	S19	shipped
Production control room	S16	shipped (optional tier)

PEG automation surface	CivicCast section	Status
Remote contribution	S17	shipped (optional tier)
Scheduled recording	S21	shipped
Custom metadata fields	S22	shipped
As-run / EPG / franchise	S23	shipped
Underwriting spots + affidavits	S24	shipped
Meeting agenda + chapters	S25	shipped
Subscription paywall	S26	backend/operator surface built; public tier discovery is not deployed, so the flow is not end-to-end usable
Analytics / audience measurement	S14	built — packaged audience reports (CSV/XML export); durable Postgres store + dashboard UI is a tracked follow-on
Accessibility WCAG 2.1 AA	S20	built — axe-core + DC-4 contrast gate (both portals); screen-reader walkthrough is manual/beta

Hardware-bounded surfaces — physical SDI, QAM modulation, EAS hardware endec, app-store publication — are out of the software boundary; the software writes the full path up to the interface, and the LPM-lab acceptance (Step 14) is where the hardware side gets proven.

Open Items And Roadmap Pointers

One external release gate remains before CivicCast can be described as a managed-service-quality claim:

- **Step 13 — release-artifact soak.** Earlier soak evidence remains historical input. A release cannot inherit exact-artifact proof from an earlier candidate; its own signed public executable must complete clean-host installation, lifecycle proof, publication, and public-download verification.
- **Step 14 — Headend acceptance at the LPM lab.** First-station beta at the LPM headend; external evidence only.

S14 and S20 closed their pinned gaps this pass, with two honestly-scoped follow-ons still open:

- **S14 — Audience reports.** `civiccast/analytics/audience_reports.py` ships franchise audience reports (per-channel totals, top assets, CSV/ XML export) over the existing aggregate analytics store. Still open, tracked separately: the durable Postgres-backed event/rollup tables (the store is still a single JSON file), the four-panel dashboard UI, a board-ready PDF, and year-over-year trend.
- **S20 — Accessibility.** axe-core CI gates plus a real DC-4 contrast gate (both portals, with an empty-scan honesty guard and a permanent falsification test) are in place. Lighthouse's accessibility category is deliberately not run as a separate CI step — it audits via axe-core internally, so the existing zero-violation axe gate is already a stricter floor. The screen-reader (NVDA/VoiceOver) walkthrough

remains a manual, beta-stage item for the master §12 release /walkthrough.

The remaining §10/§11 source modules are present, but module presence is not field proof. The stock acceptance claim remains limited to the installer and the recorded-sample rehearsal → private package → Portal approval → resident playback path described in Section A. Live ingest, 24/7 operation, station devices, and external providers require separate proof.

For deeper section reading:

- [S15 — Playout engine \(GStreamer\)](#)
- [S7 — Media lifecycle & readiness](#)
- [S8 — Health, alerting, support updates](#)
- [Legal Notices](#)
- [Patent Risk Notes](#)
- [S21 — Scheduled recording](#)
- [S25 — Meeting-agenda integration](#)
- [S26 — Subscription paywall](#)
- [S10 — Field certification & proof ladder](#)

Language Standard

All user-facing CivicCast copy follows the [Operator Language Guide](#). In short:

- **Broadcast** for the live or recorded meeting the public watches.
- **Publish** for making approved records available after review.
- **Not set up yet** or **needs IT help** instead of *blocked* when the next step is configuration.
- **Ready, check before meeting**, or **do not broadcast yet** instead of internal health-state names.
- **Auto-generated captions** and **reviewed captions** so the difference is clear to staff and to residents.